

GAP_ANALYSIS

HelixTrack Gap Analysis — 2026-06-21

Executive Summary

HelixTrack is a comprehensive JIRA alternative with multiple clients but significant implementation gaps in the core backend.

Core Backend Gaps

Database Schema (71 tables)

- **Fully Implemented:** 10 entities (14%)
- **Partially Implemented:** 4 entities (6%)
- **Not Implemented:** 57 entities (80%)

Critical Missing Features (HIGH Priority)

Entity	Status	Impact
ticket_type	✗ Missing	Required for ticket system
ticket_status	✗ Missing	Required for ticket system
workflow	✗ Missing	Referenced by projects
workflow_step	✗ Missing	Required for workflow system
board	✗ Missing	Core feature
cycle	✗ Missing	Agile feature
component	✗ Missing	Organizational feature
label	✗ Missing	Ticket tagging
priority	✗ Missing	Ticket prioritization
resolution	✗ Missing	Ticket resolution tracking
attachment	✗ Missing	File attachments
user_role	✗ Missing	RBAC
permission	✗ Missing	Access control
audit_log	✗ Missing	Activity tracking

Implemented Features (✅)

Entity	Status	Notes
project	✅ Complete	Full CRUD
ticket	✅ Complete	Full CRUD
comment	✅ Complete	Full CRUD
user	✅ Complete	Full CRUD

Web Client Gaps

Dashboard

- **Status:** Basic placeholder
- **Missing:** Real data, widgets, charts, animations

Theme System

- **Status:** CSS variables defined
- **Issue:** Inconsistent usage across components
- **Missing:** Dark mode, design token enforcement

Animations

- **Status:** Basic transitions
- **Missing:** Micro-interactions, smooth transitions, Lottie

Responsive Design

- **Status:** Desktop-focused
- **Missing:** Mobile-friendly layouts

Desktop Client Gaps

Native Integration

- **Missing:** System tray, global shortcuts, notifications
- **Missing:** File drag-and-drop, auto-updater
- **Missing:** Window state persistence

UI/UX

- **Status:** Shares code with web client
- **Missing:** Desktop-specific optimizations

Android Client Gaps

Architecture

- **Status:** Good (Hilt DI, Navigation)
- **Missing:** Full feature parity with web

UI/UX

- **Status:** Jetpack Compose
 - **Missing:** Theme consistency, animations
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iOS Client Gaps

Architecture

- **Status:** SwiftUI + GRDB
- **Missing:** Full feature parity with web

UI/UX

- **Status:** SwiftUI
 - **Missing:** Theme consistency, animations
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Cross-Platform Gaps

Design System

- **Issue:** No shared design system
- **Recommendation:** Implement OpenDesign

Theme Consistency

- **Issue:** Different implementations per platform
- **Recommendation:** Centralized design tokens

Test Coverage

- **Issue:** Inconsistent across platforms
 - **Recommendation:** Unified test strategy
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Priority Recommendations

Immediate (HIGH)

1. Implement missing core entities (ticket_type, ticket_status, workflow, board)
2. Enforce theme variables across all components
3. Add dark mode support
4. Implement real dashboard

Short-term (MEDIUM)

1. Add animations and micro-interactions
2. Implement responsive design
3. Add native desktop features
4. Create shared design system

Long-term (LOW)

1. Achieve full feature parity across clients
 2. Implement advanced analytics
 3. Add AI-powered features
 4. Create plugin system
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Cross-references

- [Architecture Overview](#)
- [UI/UX Findings](#)
- [Missing Features Report](#)

Sources verified 2026-06-22: /Volumes/T7/Projects/helix_track/core/ (sibling HelixTrack source tree) , <https://github.com/Helix-Track/Everything>

REPO-STATE-DERIVED (per §11.4.99 the sources are the cross-referenced HelixTrack source tree, following the docs/ARCHITECTURE.md precedent). Cross-referenced on 2026-06-22: - The gap analysis (“comprehensive clients but core-backend implementation gaps”) is derived from the present sibling helix_track/core/ tree vs the client trees; the cited core/MISSING_FEATURES_REPORT.md lives in that sibling tree. - The umbrella github.com/Helix-Track/Everything confirms the many-client + Core-service shape that motivates the “gaps in the core backend” thesis. - This is an assessment doc, not external instructions; no version claim to staleness-check. Re-derive the gaps from the live core/ tree as it advances.